Real Time Subband Beamformi

CB008322 (Made2)
CB011160 (AEHC)
CB011116 (RF7)
CB011159 (Made1)

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8.5.2023

Real Time Subband Beamforming features

Feature id	Release	RU family	Cell config support (CC, DL streams, subband size)
CB008322	24R2 (restricted in 23R1 and 24R1)	RF6.2 / Made2	2cc 8DL 8PRB subband size
CB011160	24R2	AEHC	1CC **16DL** 8PRB 2CC 16DL 16PRB 2CC 8DL 8PRB 3CC 16+8+8DL 16PRB
CB011116	24R3	RF7.1	1CC 16DL **4PRB** 2CC 16DL 16PRB
CNI-116383	24R2	RF7.1	1CC 8DL 8PRB (CB011116-AA)
CB011159	24R3	RF6.1 / Made1	1CC 8DL 16PRB 2CC 4DL 16PRB
CB012249	25R1	RF6.2 / Made2	1CC **16DL** 8PRB
CB011117	25R1	All above	Enhancements (More SBBF Ue's)

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Real Time Subband Beamforming

Introduction from **Feature Corner** material https://feature-corner.nsn-net.net/materials/clgglqv7f08jskz3hdz9xb8f2

Introduction

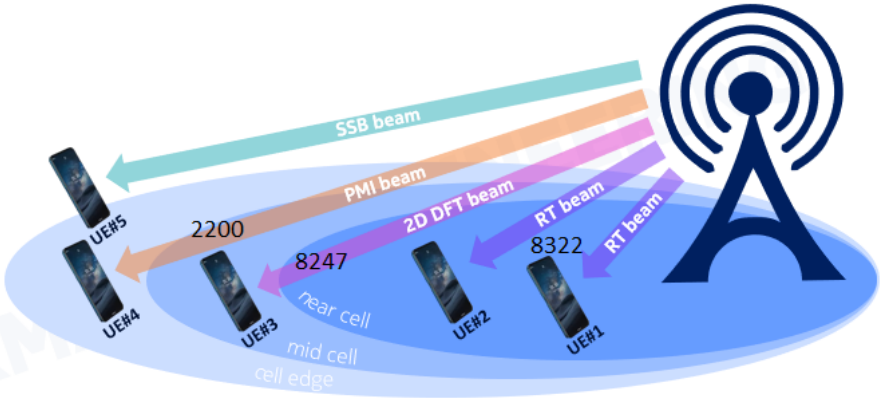
CB008322 overview

Key technologies:

- This feature supports **sounding-based DL short-term sub-band beamforming** for NR FR1-TDD eCPRI-based massive MIMO.
- **Dynamic fallback** to legacy beamforming schemes.

Customer benefits:

- This feature **will improve DL throughput performance**.
- UEs that are scheduled with real time beamforming will be able **to reach peak throughput more often**.
- The nominal **peak throughput is not changed**.



Different UEs - based on data buffer and radio conditions - **can use Real-Time (RT) beams** or **2D DFT/PMI /SSB beams**

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Real Time Subband Beamforming

Introduction

CB008322 in a nutshell (2/3)

- The CB008322 feature supports **fallback** to one of **legacy beamforming** algorithms with **RRC re-configuration** to sub-band sounding **or periodic sounding de-activation** as well as **dynamic beamforming algorithm fallback** without RRC re-configuration.
- The UE can be scheduled **either with sub-band beamforming algorithm or with CB008247 algorithm** per-slot basis for all PDSCH PRBs allocated to this UE when configured with full-band antenna switching sounding, depending on the sounding signal quality.
- The CB008322 feature supports **frequency domain multiplexing scheduling between sub-band beamforming users and legacy beamforming users** (SU-MIMO and/or MU-MIMO up to 8 layers) per slot basis.
- There is support for **basic capacity target 16 beamforming PRB granularity** and finer depending on BB/RU cell-sets.
- The CB008322 is applicable to **64TRX**, **32TRX products with NDM sounding capability** and to baseband deployment platforms: **ABIO/ASOE/ABIN/ABIP/ABIQ**.

New fallback mechanism **with** RRC reconfiguration

Dynamic fallback **without** RRC reconfiguration

Sub-band BF UEs can be FDMed with UEs using legacy BF methods

64/32 TRX RU support

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Real Time Subband Beamforming

Introduction

CB008322 in a nutshell (3/3)

This feature **supports intra-gNB intra & inter-band CA** for the following cases:

- **Case 1: TDD + TDD DL CA**, sounding-based short-term sub-band beamforming is applied to PCell
- **Case 2: FDD + TDD DL CA**, sounding unavailable on SCell as no support of carrier switching sounding
- **Case 3: TDD + FDD DL or UL CA**, sounding-based short-term sub-band beamforming is applied to PCell and **can be managed by O&M parameters** for **critical band combo** cases
- **Case 4: FDD + TDD DL + UL CA**, sounding-based short-term sub-band beamforming is not applied to SCell.

Inter-gNB CA **(as introduced by** CB008140 Inter-gNB DL carrier aggregation FR1-FR1**)** is **not supported** with CB008322.

Intra-gNB intra & inter-band Carrier Aggregation support

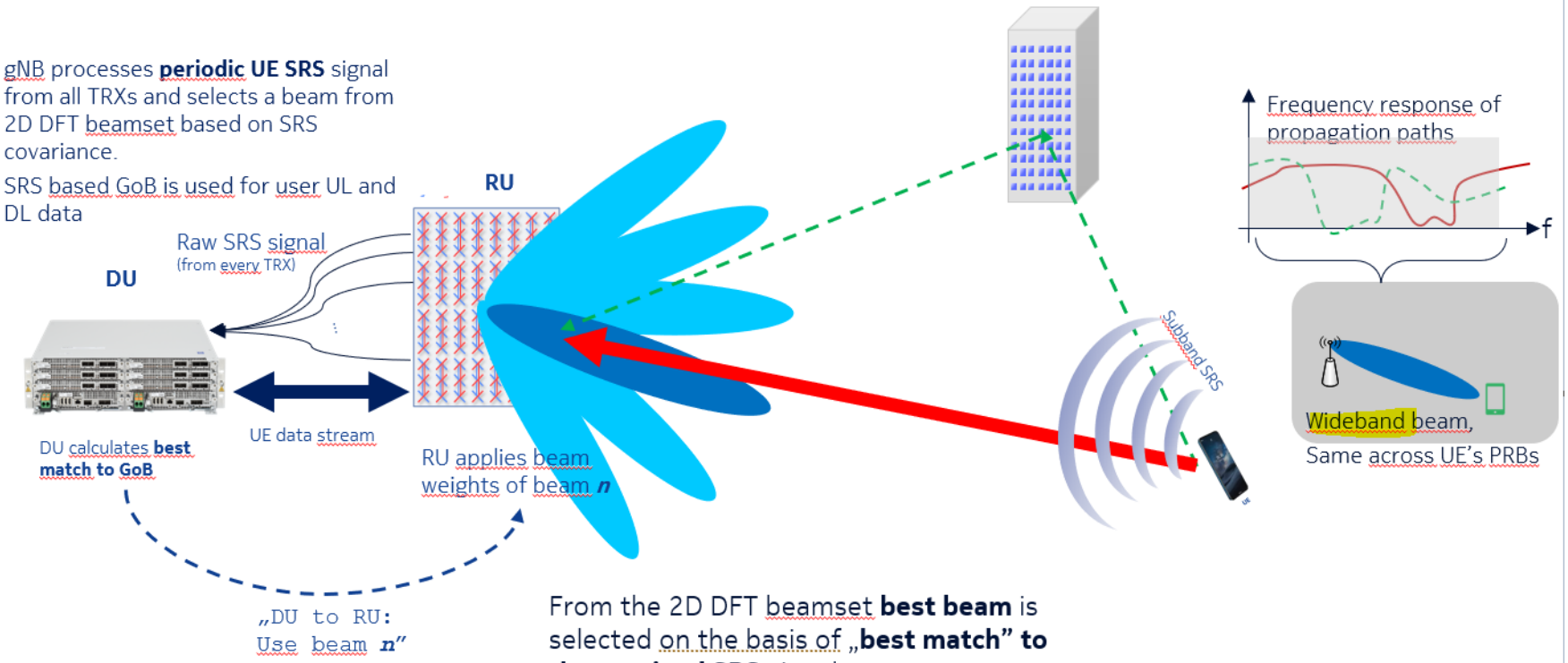
Inter-gNB Carrier Aggregation **not supported**

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Beamforming evolution

SRS based beamforming: Grid of Beams (GoB) DLSRS BF / 8247

- gNB processes **periodic UE SRS** signal from all TRXs and selects a beam from 2D DFT beamset based on SRS covariance.
- SRS based GoB is used for user UL and DL data



From the 2D DFT beamset **best beam** is selected on the basis of „**best match" to the received SRS** signals.

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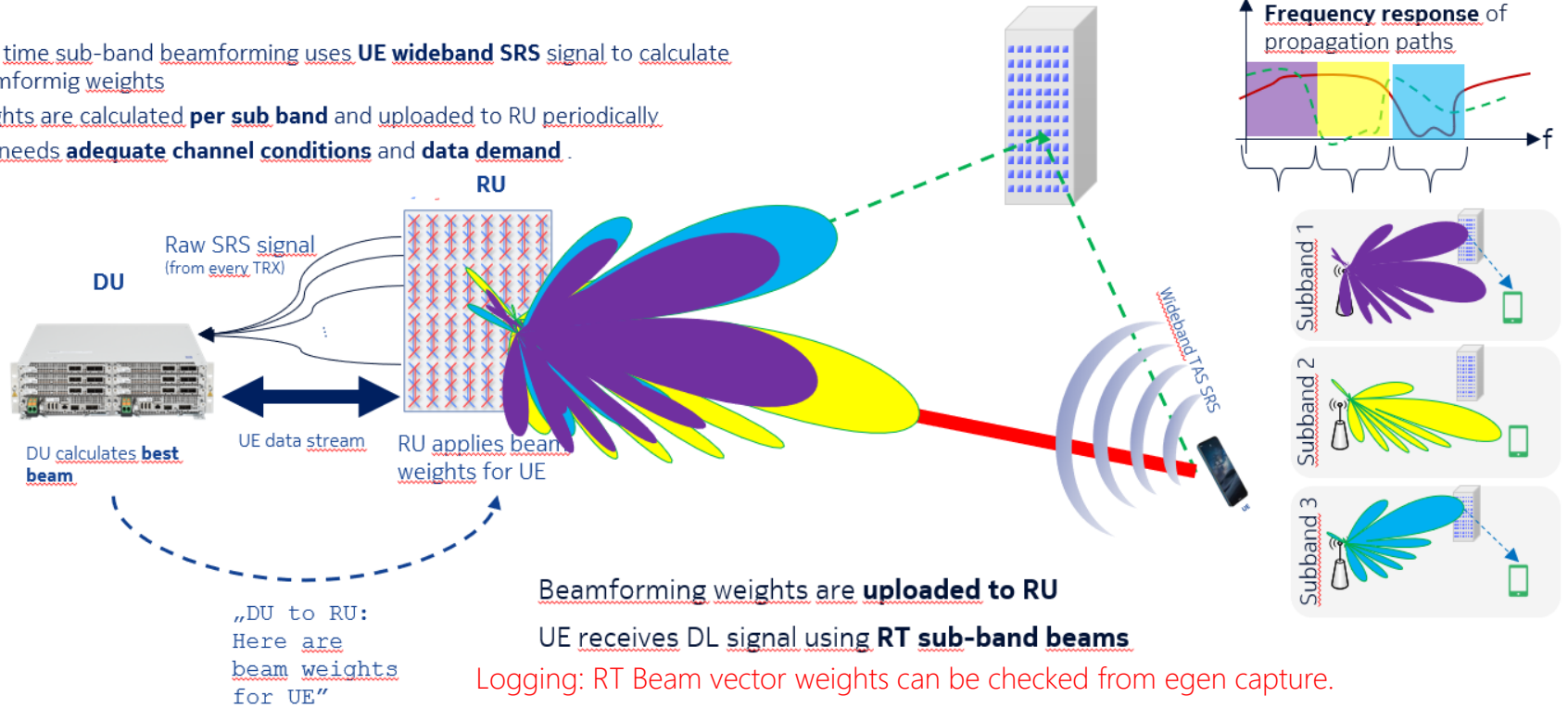
Subband size vs num of subbands 100Mhz (273PRB) Cell BW
Subband size 272 -> wideband
Subband size 8 -> 34 subbands (Max in RF6.2 and AEHC)
Subband size 4 -> 68 subbands (RF7)

Beamforming evolution

animation

Real time (RT) sub-band beamforming

- Real time sub-band beamforming uses **UE wideband SRS** signal to calculate beamformig weights
- Weights are calculated **per sub band** and uploaded to RU periodically
- UEs needs **adequate channel conditions** and **data demand** .



Beamforming weights are **uploaded to RU**

UE receives DL signal using **RT sub-band beams**

Logging: RT Beam vector weights can be checked from egen capture.
(Some L1 snapshot functionality should be possible also)

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Feature Design Details

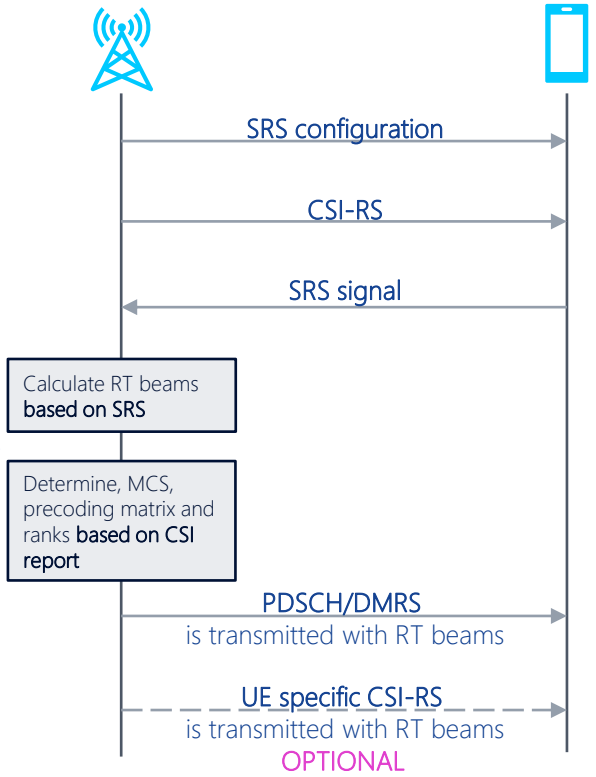
SRS based BF processing chain overview

In high level, the processing chain is shown below:

- **UE/Cell specific CSI-RS is configured to UE** during UE setup.
- gNB monitors **UE's channel condition and buffer status**
- gNB **configures wideband SRS resource to UE via RRC reconfiguration** if channel condition is good and buffer is high.
- **UE transmits SRS** based on gNB configuration
- **gNB calculates the RT beam(s) for PDSCH/DMRS transmission**, UE specific CSI-RS transmission.
- **gNB determines the MCS, precoding matrix and ranks based** on CSI report.
- **PDSCH/DMRS is transmitted** with the RT beam(s)
- **UE specific 4 ports CSI-RS if configured is transmitted** with four wideband RT beams.

After CB008322 two kinds of algorithms are supported. If actDlSrsBm is **true**, **Long-term algorithm** will be enabled.

If **actDlSrsSubbandBm** is true, **short-term algorithm** will be enabled together with long-term algorithm.

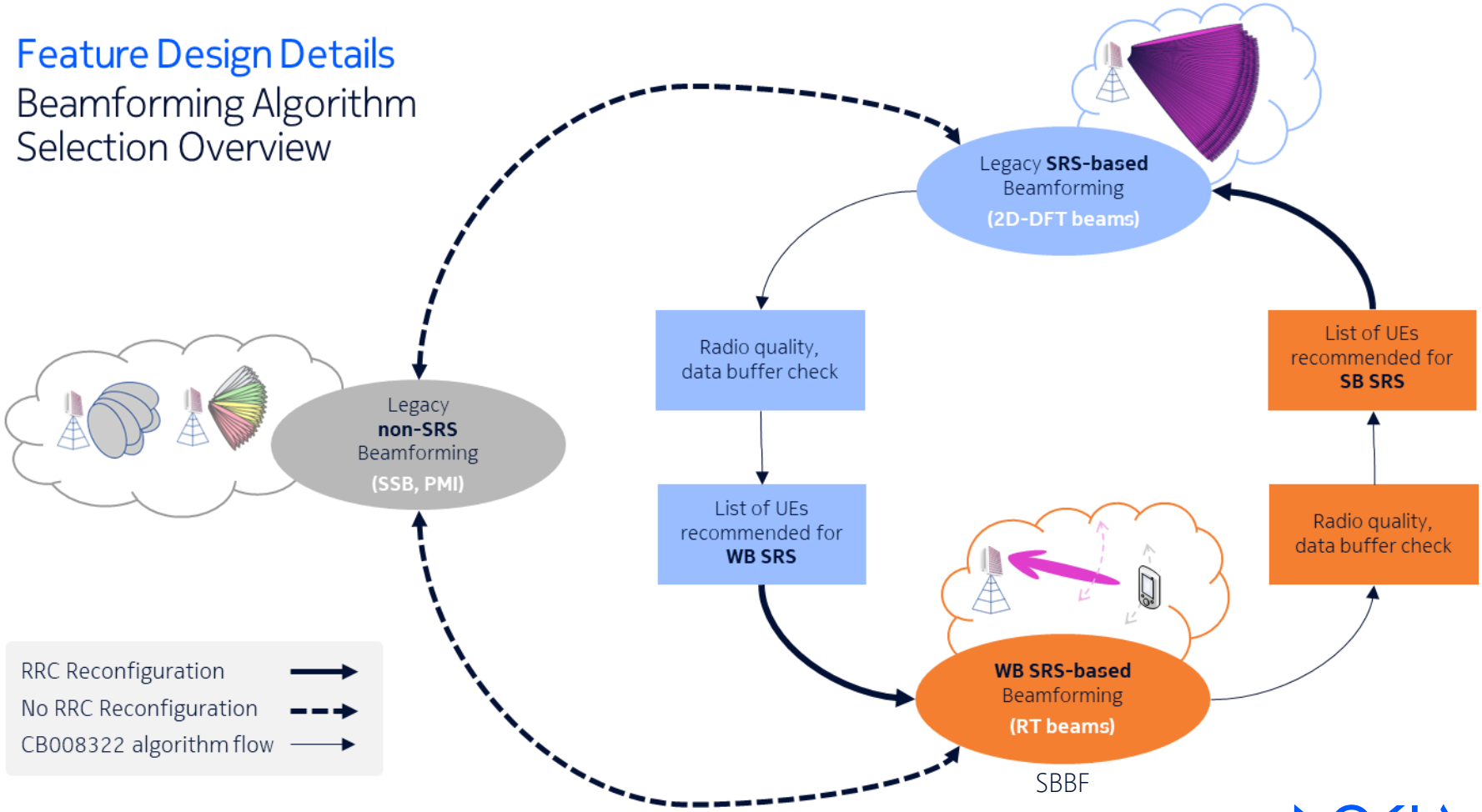


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Feature Design Details

Beamforming Algorithm Selection Overview



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Confusing Terms Subband Beamforming and wideband SRS

Sometimes terms gets mixed up here when we speak **Subband beamforming** which configures Ue to use **wideband TAS SRS**.

Where as GoB BF 8247 is wideband BF with subband TAS SRS.

SRS BF feature	Called as	TAS SRS config for Ue	beamtype in ttiTrace	srsState in ttiTrace
CB008247	DL SRS/wideband BF / GoB BF	subband SRS	DFTNONTAPERING DFTTAPERING (MuMIMO)	sbSRS
8322/11160/11116/11159	**RT Subband BF**	**wideband SRS**	SUBBAND	wbSRS





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Configuring Real Time Subband Beamforming

For configuring SBBF there is a lot of new parameters please check how to page for easy start here:
HowTo: Configure sub-band beamforming in WebEM - RF Verification Unit / Beamforming - Dashboard (nokia.com)

Feature activation:
Set actDlSrsSubbandBm=true
Add srsForSubbandBm structure
Add wbSbSrsSwitch structure
Set cbcTasConflictPolicy =TAS SRS first

RnD flags:
For beam measurement BWV update can be stopped with NrUlPhy flag enableBWVSrsFhResp (0x800063=0) after Ue attach and data transfer start.
(removed from webEm need to check if working via aashell)

Parameters introduced in 23R4, 24R1 and 24R2:
NRBTS/NRCELL-actDlSrsSubbandBm
NRBTS/NRCELL-srsForSubbandBm
NRBTS/NRCELL-srsForSubbandBm-bfSubbandSize
NRBTS/NRCELL-srsForSubbandBm-maxNumUeSrs
NRBTS/NRCELL-srsForBeamMgmt-comb
NRBTS/NRCELL-srsForSubbandBm-actTasSrsForCriticalBandCombo
NRBTS/NRCELL-srsForSubbandBm-bfSubbandSizeOutput
NRBTS/NRCELL-srsForSubbandBm-cbcTasConflictPolicy
NRBTS/NRCELL-srsForSubbandBm-dynamicSrsRmMode
NRBTS/NRCELL-srsForSubbandBm-srsPowerThreshold (replacing srsQualityThreshold)
NRBTS/NRCELL-srsForSubbandBm-srsSinrThreshold
NRBTS/NRCELL-srsForSubbandBm-srsUePeriodicityInput
NRBTS/NRCELL-srsForSubbandBm-srsUePeriodicityOutput
NRBTS/NRPMRNL-miNrBeamforming2
NRBTS/NRPMRNL-miNrBfConfig
NRBTS/NRPMRNL-miNrSrsQuality
NRBTS/NRCELL-wbSbSrsSwitch
NRBTS/NRCELL-wbSbSrsSwitch-channelStabilityHysteresis
NRBTS/NRCELL-wbSbSrsSwitch-channelStabilityThreshold
NRBTS/NRCELL-wbSbSrsSwitch-enableSbSrsVolatilityCheck
NRBTS/NRCELL-wbSbSrsSwitch-maxWbSbSwitch
NRBTS/NRCELL-wbSbSrsSwitch-numWbSrsConfigSample
NRBTS/NRCELL-wbSbSrsSwitch-numWbSrsReleaseSample
NRBTS/NRCELL-wbSbSrsSwitch-prbRatioHysteresis
NRBTS/NRCELL-wbSbSrsSwitch-prbRatioTreshold
NRBTS/NRCELL-wbSbSrsSwitch-prbSampleHighLowPercentage
NRBTS/NRCELL-wbSbSrsSwitch-rankHysteresis
NRBTS/NRCELL-wbSbSrsSwitch-rankTreshold
NRBTS/NRCELL-wbSbSrsSwitch-rlqBadPercentage
NRBTS/NRCELL-wbSbSrsSwitch-rlqGoodPercentage
NRBTS/NRCELL-wbSbSrsSwitch-seHysteresis
NRBTS/NRCELL-wbSbSrsSwitch-seThreshold
NRBTS/NRCELL-wbSbSrsSwitch-sortingMetric
NRBTS/NRCELL-wbSbSrsSwitch-wbSrsConfigProhibitTimer
NRBTS/NRCELL-wbSbSrsSwitch-wbSrsPowerHysteresis
NRBTS/NRCELL-wbSbSrsSwitch-wbSrsPowerTreshold
NRBTS/NRCELL-wbSbSrsSwitch-wbSrsReleaseProhibitTimer

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Configuring Real Time Subband Beamforming

Some notes for parametrization:

NRBTS/NRCELL-srsForSubbandBm-**bfSubbandSize** = 4, 8, 16… Refer to supported size depending on cell configs and try to use smallest allowed value (wideband can be used also for some measurement/investigation purposes).

NRBTS/NRCELL-srsForSubbandBm-**maxNumUeSrs**: 1-4, could have effect on TAS srs periodicity configuration.

If **srsUePeriodicityInput** value is bigger than allowed -> OAM calculates minimum allowed value and then configured value is showed in srsUePeriodicityOutput

slowPrbPooling	maxNumUeSrs	frame structure	has muted special slots in a frame?	min wideband SRS periodicity (slots)
disabled	<3	1208/1070	yes	40
			no	20
		1116	no	40
	>=3	1208/1070	yes	80
			no	40
		1116	no	80
enabled	<3	1208/1070	yes	80
			no	40
		1116	no	80
	>=3	1208/1070	yes	160
			no	80
		1116	no	160

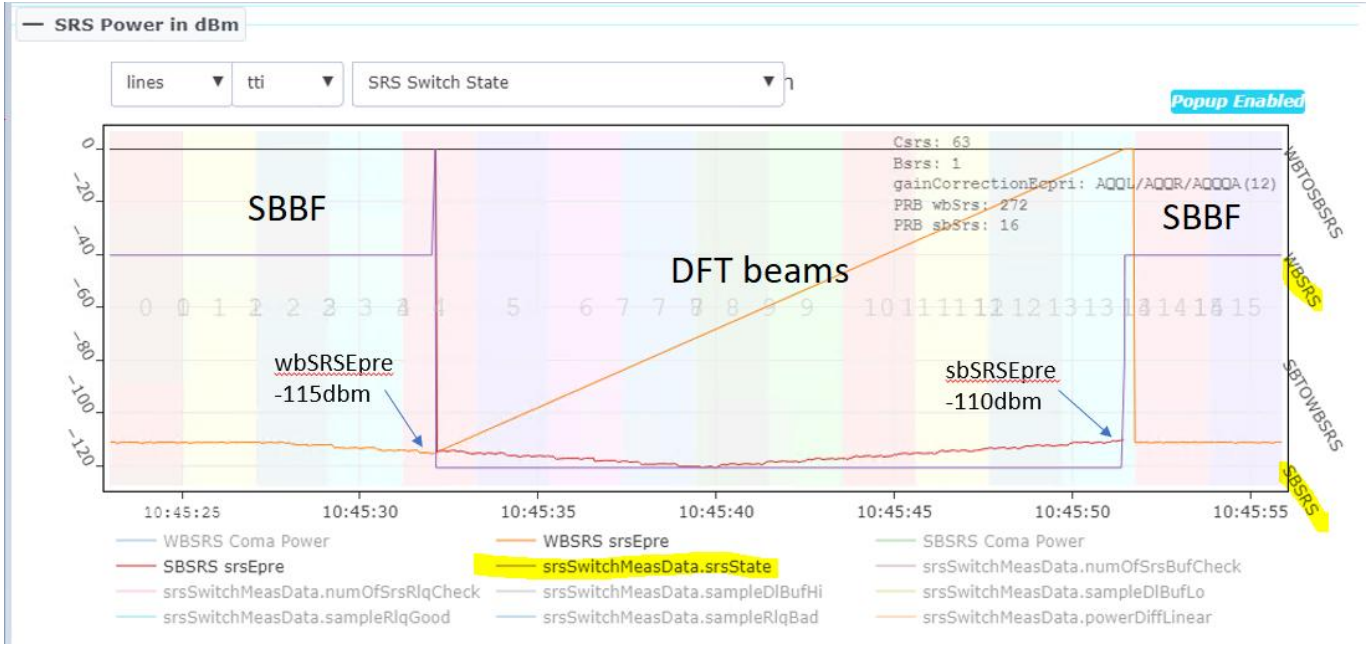
Parameters introduced in 23R4, 24R1 and 24R2:
NRBTS/NRCELL-actDlSrsSubbandBm
NRBTS/NRCELL-srsForSubbandBm
NRBTS/NRCELL-srsForSubbandBm-bfSubbandSize
NRBTS/NRCELL-srsForSubbandBm-maxNumUeSrs
NRBTS/NRCELL-srsForBeamMgmt-comb
NRBTS/NRCELL-srsForSubbandBm-actTasSrsForCriticalBandCombo
NRBTS/NRCELL-srsForSubbandBm-bfSubbandSizeOutput
NRBTS/NRCELL-srsForSubbandBm-cbcTasConflictPolicy
NRBTS/NRCELL-srsForSubbandBm-dynamicSrsRmMode
NRBTS/NRCELL-srsForSubbandBm-srsPowerThreshold (replacing srsQualityThreshold)
NRBTS/NRCELL-srsForSubbandBm-srsSinrThreshold
NRBTS/NRCELL-srsForSubbandBm-srsUePeriodicityInput
NRBTS/NRCELL-srsForSubbandBm-srsUePeriodicityOutput
NRBTS/NRPMRNL-miNrBeamforming2
NRBTS/NRPMRNL-miNrBfConfig
NRBTS/NRPMRNL-miNrSrsQuality
NRBTS/NRCELL-wbSbSrsSwitch
NRBTS/NRCELL-wbSbSrsSwitch-channelStabilityHysteresis
NRBTS/NRCELL-wbSbSrsSwitch-channelStabilityThreshold
NRBTS/NRCELL-wbSbSrsSwitch-enableSbSrsVolatilityCheck
NRBTS/NRCELL-wbSbSrsSwitch-maxWbSbSwitch
NRBTS/NRCELL-wbSbSrsSwitch-numWbSrsConfigSample
NRBTS/NRCELL-wbSbSrsSwitch-numWbSrsReleaseSample
NRBTS/NRCELL-wbSbSrsSwitch-prbRatioHysteresis
NRBTS/NRCELL-wbSbSrsSwitch-prbRatioTreshold
NRBTS/NRCELL-wbSbSrsSwitch-prbSampleHighLowPercentage
NRBTS/NRCELL-wbSbSrsSwitch-rankHysteresis
NRBTS/NRCELL-wbSbSrsSwitch-rankTreshold
NRBTS/NRCELL-wbSbSrsSwitch-rlqBadPercentage
NRBTS/NRCELL-wbSbSrsSwitch-rlqGoodPercentage
NRBTS/NRCELL-wbSbSrsSwitch-seHysteresis
NRBTS/NRCELL-wbSbSrsSwitch-seThreshold
NRBTS/NRCELL-wbSbSrsSwitch-sortingMetric
NRBTS/NRCELL-wbSbSrsSwitch-wbSrsConfigProhibitTimer
NRBTS/NRCELL-wbSbSrsSwitch-wbSrsPowerHysteresis
NRBTS/NRCELL-wbSbSrsSwitch-wbSrsPowerTreshold
NRBTS/NRCELL-wbSbSrsSwitch-wbSrsReleaseProhibitTimer

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Configuring Real Time Subband Beamforming

Under NRBTS/NRCELL-**wbSbSrsSwitchtch** there are multiple parameters for triggering **sbSRS <-> wbSRS** configuration for Ue. E.g. thresholds for wbSRSpower, -SINR, PRB ratio or spectral efficiency.

You may need to adjust **wbSRSpowerThreshold** to get UE configured with wbSRS (default -101 to e.g -110) in you test environment depending on path loss. Here is example from path loss sweep test (antenna path attenuation 10-20-10db 1db/sec) where wbSRS->sbSRS switch happens on -115dbm and recovery on -110dbm



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Configuring Real Time Subband Beamforming

And here are those fallback and recovery triggering points in Emil DL dataview from same test log (ttiTrace).

wbSRS <-> sbSRS power thresholds
srsPowerThreshold = -130
wbSrsPowerThreshold = **-110**
wbSrsPowerHysteresis = 5

-> wbSRS to sbSRS
wbSRSEpre= -110 - 5 => **-115 dbm**
(L2 from Coma)

-> sbSRS to wbSRS
sbSRSEpre = -110 dbm
(L2 from Coma)

srsPowerThreshold = -130
This is used by L1 (bip port powers, Not Coma) can trigger wbSRS->sbSRS too, but not obvius due below nidd rules

				srsEpreLinear	SB-SRS EPRE	srsEpreLinearThd
WBSRS	100	2963	9764	6282.772949	-115.218486	6606.937012
WBSRS	100	2963	9764	6282.772949	-115.218486	6606.937012
WBSRS	100	2963	9764	6282.772949	-115.218486	6606.937012
WBSRS	100	2963	9759	6428.146973	-115.119142	6606.937012
WBSRS	100	2963	9759	6428.146973	-115.119142	6606.937012
WBTOSBSRS	0	1	0	0	-Infinity	0
WBTOSBSRS	0	1	0	0	-Infinity	0
WBTOSBSRS	0	1	0	0	-Infinity	0
SBSRS	0	1	0	0	-Infinity	0
SBSRS	0	2865	9759	0	-Infinity	0
SBSRS	86	2222	9759	0	-Infinity	0
SBSRS	76	1171	9759	0	-Infinity	0
SBSRS	79	1171	9759	0	-Infinity	0
SBSRS	75	1171	9759	0	-Infinity	0
SBSRS	84	1171	9759	10038.436523	-113.183339	20892.96875
SBSRS	100	1264	9759	10038.436523	-113.183339	20892.96875
SBSRS	98	1768	9752	21246.013672	-109.927225	20892.96875
SBSRS	98	1667	9752	21246.013672	-109.927225	20892.96875
SBSRS	96	1768	9752	20816.234375	-110.015978	20892.96875
SBTOWBSRS	0	1	0	0	-Infinity	0
SBTOWBSRS	0	1	0	0	-Infinity	0
SBTOWBSRS	0	1	0	0	-Infinity	0
WBSRS	0	1	-10000	0	-Infinity	0
WBSRS	0	1212	-10000	0	-Infinity	0

wbSRSEpre
-115dbm

sbSRSEpre
-110dbm

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Configuring Real Time Subband Beamforming

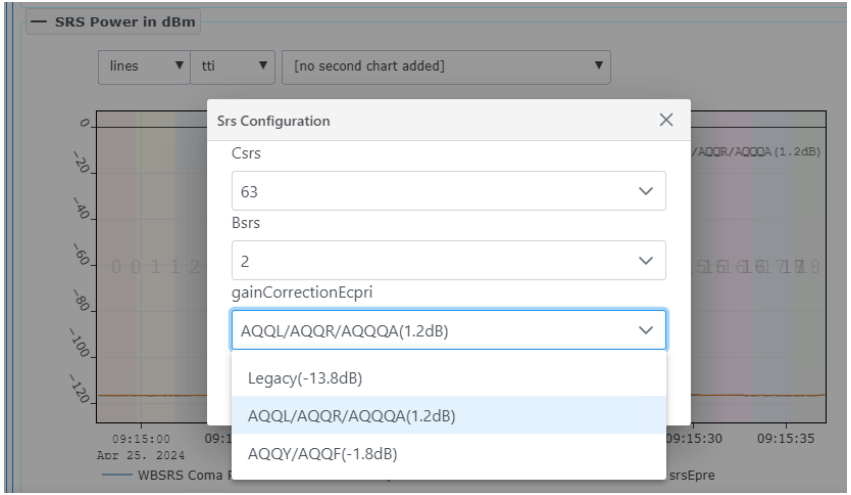
Received wbSRSpower is shown in ttiTrace in linear values and tools can calculate dbm value, but need to add correct gainCorrectionEcpri value for tools. Made2 RU's are exception here as they use different values than other RU's(due CB010765 eCPRI UL Scaling change for RF6.2 MAA Radios)

Made2 64TRX (AQQL) gainCorrectionEcpri = 1.2 (12 in L2 cell setup msg).

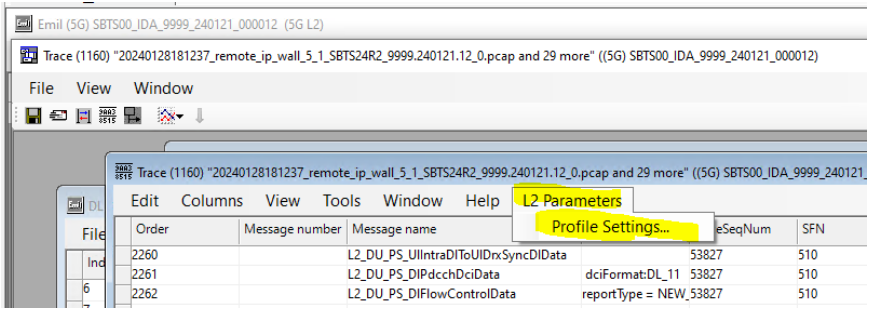
Made2 32TRX (AQQY) gainCorrectionEcpri = -1.8. (-18 in L2 cell setup msg).

All other Ru's (e.g AVQL, AVQC and AEHC) use gainCorrectionEcpri =-13.8 (-138 in L2 cell cetup msg).

UpAnalyzer setting in SRS power window



In Emil Trace view and source L2 Parameters



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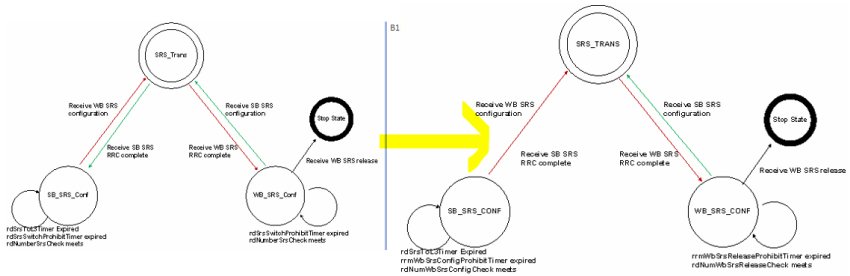
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Configuring Real Time Subband Beamforming

For more detailed info about other wbSRS <-> sbSRS (RT SBBF <-> GoB BF) switch triggers please check CFAM or this one note page Dynamic RRC of SB and WB (Web view)

Dynamic RRC of SB and WB

Thursday, August 10, 2023 4:59 PM



Dynamic SRS reconfiguration in RRC

SB SRS→WB SRS			Default
rrmMetricBufferHi and	>=	NIDD:wbSbSrsSwitch/prbSampleHighLowPercentage	60%
rrmMetricRlqGood and	>=	wbSbSrsSwitch/rlqGoodPercentage	20%
And		UE is under SRS_DL_BM_VALID	
And		UE is not 'WB SRS constraint UE'	
And		channel volatile check for this UE succeeds	
And		rdSrsToL3Timer(cell specific, start from cellsetup) is expired	500 slots
WB SRS→SB SRS			
rrmMetricBufferLo or	>=	NIDD:wbSbSrsSwitch/prbSampleHighLowPercentage	60%
rrmMetricRlqBad or	>=	NIDD:wbSbSrsSwitch/rlqBadPercentage	50%
rrmRtBeamNewTxNackRatio or	>=	rdRtBeamHarqNackPercentage	100%
Or		rdCqiExpiryTime (5G_L2_1026) expired	
Or		**rrmCoMaAgedTimer** for DL BM expires (5G_UP_ALG_6965) for UEs under **SRS_DL_BM_VALID status**	
Or		UE has high percentage of BWV failure (BIP)	>=40%

Checks per sample

Good RLQ			Default
spectral efficiency and	>=	NIDD:wbSbSrsSwitch/seThreshold	9.1
Rank and	>=	NIDD:wbSbSrsSwitch/rankThreshold	3
UL SRS power and	>=	NIDD:wbSbSrsSwitch/wbSrsPowerThreshold, from CoMa	-101dBm
averaged correlation	>=	NIDD:wbSbSrsSwitch/channelStabilityThreshold	0.3
Bad RLQ			
spectral efficiency or	< rrmSbSrsSeTreshold	max {NIDD:wbSbSrsSwitch/seThreshold – wbSbSrsSwitch/seHysteresis, 0}	9.1-5=4.1
Rank or	< rrmSbSrsRankTresh	max {NIDD:wbSbSrsSwitch/rankTreshold - wbSbSrsSwitch/rankHysteresis, 1}	3-1=2
UL SRS power or	< rrmSbSrsQualTresh	NRCELL/wbSbSrsSwitch.wbSrsPowerThreshold - NRCELL/wbSbSrsSwitch.wbSrsPowerHysteresis	-101-5=-106dBm
averaged correlation	< rrmSbSrsCorrTresh	max { wbSbSrsSwitch/channelStabilityThreshold - wbSbSrsSwitch/channelStabilityHysteresis , 0}	0.3-0.1=0.2
SINR	< rrmSbSrsSinrTresh	NRCELL/srsForSubbandBm.srsSinrThreshold + 2dB	Not enabled

BufferHi			Default
averaged scheduled PRB ratio in percentage	> rrmWbSrsPrbThreshold	NIDD:wbSbSrsSwitch/prbRatioThreshold	20%
BufferLo			
averaged scheduled PRB ratio in percentage	< rrmSbSrsPrbThreshold	{ wbSbSrsSwitch/prbRatioThreshold-wbSbSrsSwitch/prbRatioHysteresis, 0}	20-15=5%

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GoB BF vs RT SBBF when spatially separated paths

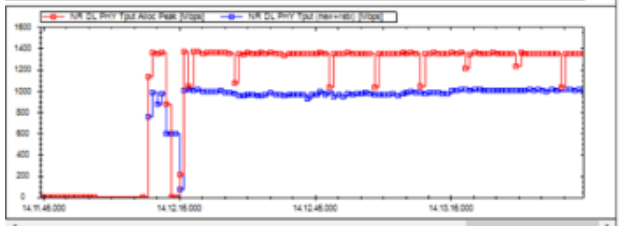
In GoB BF (e.g. 8247) algorithm tries always to find out two best beam directions for four layers and data is transferred wth those two patternId's from two polarzations.

In RT SBBF (8322) all four layers have independent direction and polarization

8247 DL t-put 995Mbps

8247

Item	DL Configuration	Run-Time Value
DL Modulation	256QAM	256QAM
DL MIMO Layer	4	3,8 (avg.)
DL PRB Number	273	273,0 (avg.)
DL Assign Number	1600/sec (TDD 30k: DDDSSSSSSSSSSSS...	1425/sec
DL Slot PDSCH SLIV	14 sym(w/ PDCCH)	13,00 sym (avg.)
Special Slot PDSCH SLIV	DL part tdd0 10 sym, tdd1 0 sym	11,00 sym (avg.)
Throughput [Mbps]	1353,56 (peak)	994,56 (73,5% of peak)
BLER [%]	10% (typical)	6,0%

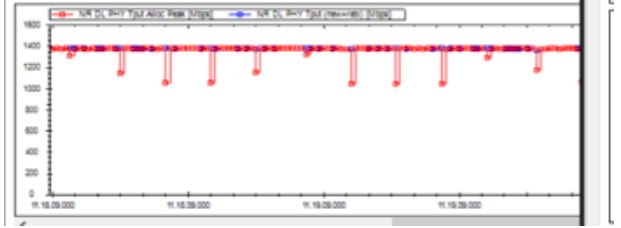


8322 DL Tput 1375Mbps

GAIN = 38%

8322 (subband size =272)

Item	DL Configuration	Run-Time Value
DL Modulation	256QAM	256QAM
DL MIMO Layer	4	4,0 (avg.)
DL PRB Number	273	273,0 (avg.)
DL Assign Number	1600/sec (TDD 30k: DDDSSSSSSSSSSSS...	1423/sec
DL Slot PDSCH SLIV	14 sym(w/ PDCCH)	13,00 sym (avg.)
Special Slot PDSCH SLIV	DL part tdd0 10 sym, tdd1 0 sym	11,00 sym (avg.)
Throughput [Mbps]	1374,80 (peak)	1375,18 (100,0% of peak)
BLER [%]	10% (typical)	0,0%

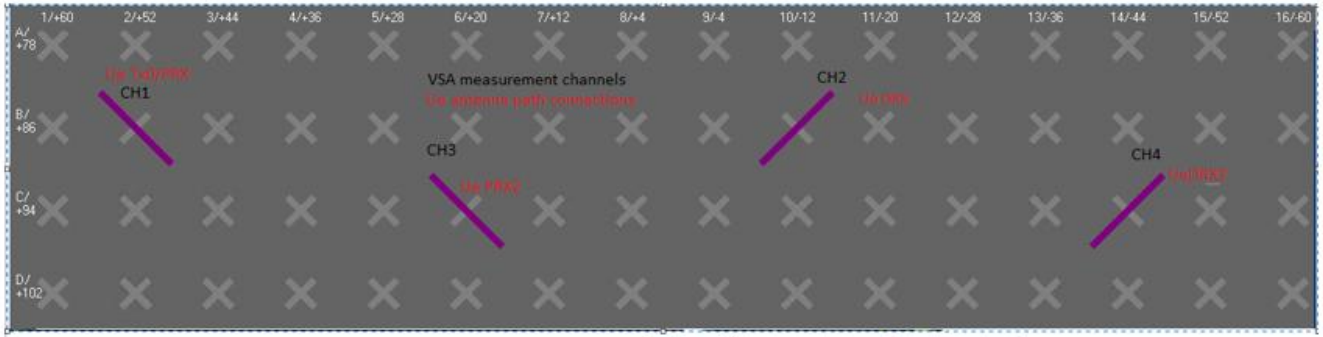


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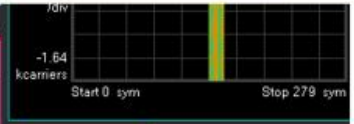
RT Subband Beamforming when spatially separated paths

In this VSA DMRS port measurement we can see that only one DMRS port is having good power and EVM level per measurement channel.



D: DMRS MIMO Info

Name	Meas. Channel	Antenna Port	Despread EVM (%rms)	Power (dBm)	TAE (sec)	Frequency Offset (Hz)	Phase Offset (deg)
PDSCH0_DMRS	Ch1	Port0	75.081	-101.62	0	-254.4	11.24
PDSCH0_DMRS	Ch1	Port1	0.778	-65.01	-1.36344 µ	39.595	50.37
PDSCH0_DMRS	Ch1	Port2	29.209	-96.46	-1.25882 µ	-174.398	-4.72
PDSCH0_DMRS	Ch1	Port3	25.074	-94.25	-1.42522 µ	-366.913	-11.15
PDSCH0_DMRS	Ch2	Port0	0.786	-65.66	-1.35915 µ	80.9308	74.10
PDSCH0_DMRS	Ch2	Port1	37.220	-97.94	-1.34011 µ	-460.044	-24.14
PDSCH0_DMRS	Ch2	Port2	22.278	-93.76	-1.42953 µ	-227.067	20.58
PDSCH0_DMRS	Ch2	Port3	35.061	-97.01	-1.4047 µ	-353.365	0.80
PDSCH0_DMRS	Ch3	Port0	48.311	-104.71	-1.60615 µ	-461.205	9.24
PDSCH0_DMRS	Ch3	Port1	8.000	-80.28	-1.37493 µ	-252.835	-52.40
PDSCH0_DMRS	Ch3	Port2	1.789	-60.73	-1.34633 µ	105.784	-172.27
PDSCH0_DMRS	Ch3	Port3	123.648	-100.95	1.32726 µ	-289.318	-7.13
PDSCH0_DMRS	Ch4	Port0	28.498	-91.56	-1.43067 µ	-193.703	-33.31
PDSCH0_DMRS	Ch4	Port1	28.314	-91.30	-1.36498 µ	-451.644	-25.07
PDSCH0_DMRS	Ch4	Port2	74.318	-91.00	-1.36585 µ	-512.117	0.29
PDSCH0_DMRS	Ch4	Port3	2.480	-61.81	-1.34869 µ	85.3921	74.63



F: Frame Summary

Name	EVM (%rms)	SINR (dB)	Power per RE (dBm)	Modu
PSS	1.689	35.45	-71.34	
SSS	1.672	35.54	-71.32	
PBCH	2.592	31.73	-71.35	
PBCH_DMRS	1.851	34.65	-71.35	
PDSCH0	3.754	28.51	-59.80	QA
PDSCH0_DMRS	2.635	31.58	-62.24	
PDSCH0_Layer0	1.871	34.56	***	QA
PDSCH0_Layer1	1.695	35.42	***	QA
PDSCH0_Layer2	4.455	27.02	***	QA
PDSCH0_Layer3	5.495	25.20	***	QA

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RT Subband Beamforming when spatially separated paths

Plotting RT beam weights with Metsa

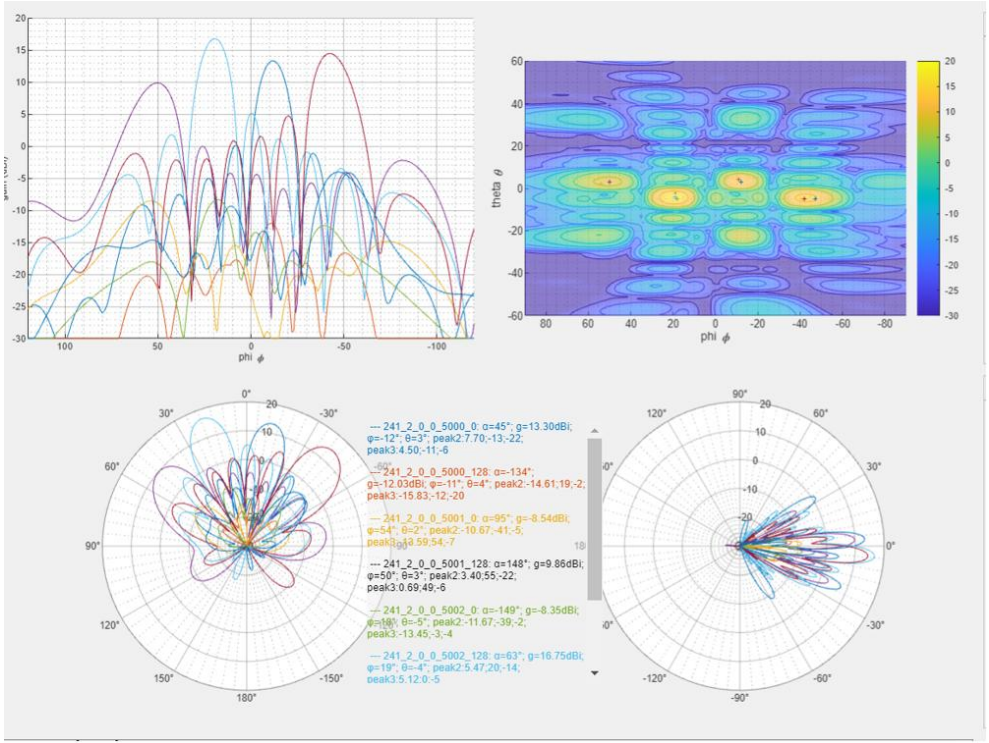
1. Take eGen capture (FCP included)

2. Open Metsa use 3.8.3 version or newer. DOWNLOAD and install to your pc from:
mMIMO & Beamforming - Metsa - All Documents (sharepoint.com)
OR
\\eseefsn50.emea.nsn-net.net\rotta4internal\5G_3\oulu_l3_camp\TheWall5\tools\metsa

3. Use eGen tab and load egen capture file there. Then select e.g first 8 beams to plot for 4 layers (one BWV update for one subband).

4. Go to Polku tab (will use egen data) select eGen RT-Bwv /Ch-Est as input and select 2 beams to plot polarization data.

please check Metsa tool guide with pictures if needed.
Metsa guide

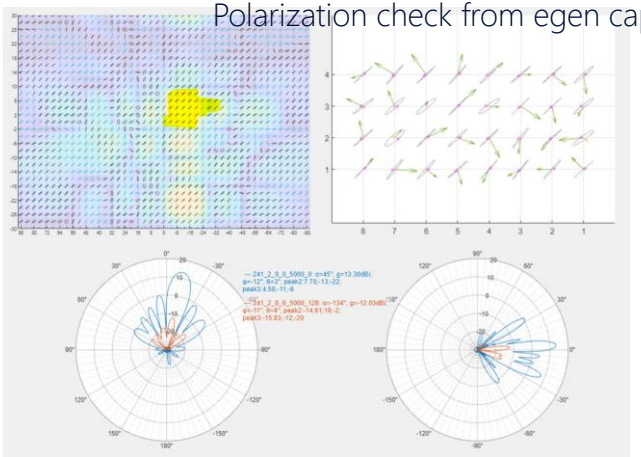


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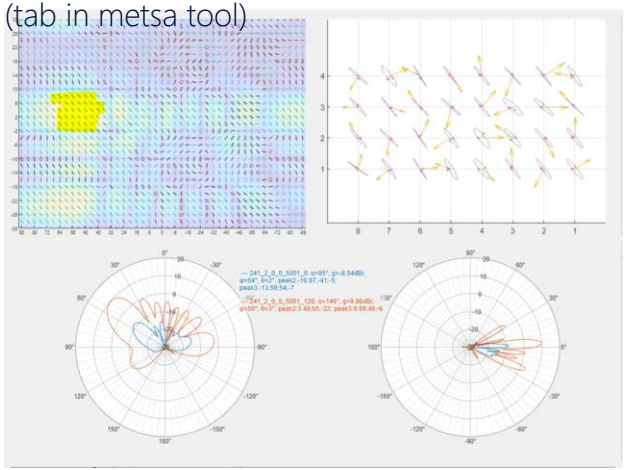
RT Subband Beamforming when spatially separated paths

Polarization check from egen capture with polku (tab in metsa tool)

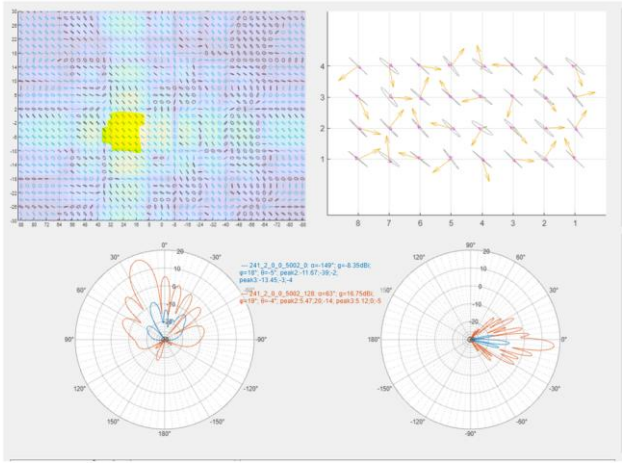
Layer#1



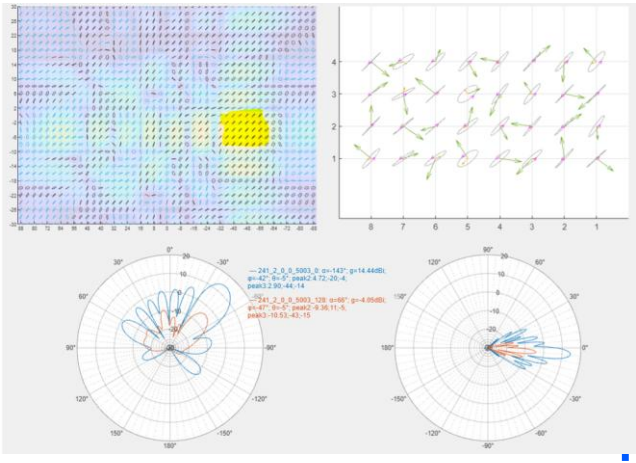
Layer#2



Layer#3



Layer#4



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RT SBBF (8322) performance vs DL SRS BF (8247) on Fading channel

Our fading channel testing result are close to simulated results even we can not use all the same parameters exactly. Single User DL t-put has gain with RT SBBF vs DLSRS BF in slow speed scenario at 3kmh (Simulated +10%).

In 15kmh there is a small loss already (Simulated -1.8%).

MMSE Performance Gain

Click to edit headline

Simulated results

SRS(ms)/Speed(km/h)	5(15)	5(15)	10(3)	10(15)	10(15)	20(3)	20(15)	20(30)
BF subband (PRB)	4	8	8	4	8	8	8	8
Avg Cell Thp (kbps)	801992	801687	910453	795045	795098	839029	785921	784563
MMSE % to 8247	-1.2%	-1.2%	10.2%	-1.8%	-1.8%	1.8%	-2.7%	-2.7%

Channel Type	LOS / NLOS
Wireless Channel	Fading channel : 2 / 6 clusters with 20 rays
Cluster Delay Spread	LOS (30/93 wo Omg) , NLOS (100/364)
UE Antenna Model	PIFA / Nokia Reference (Dipole / S20)
UE Antenna	2T4R / 4T4R
SRS Periodic	5ms / 10ms / 20ms
CSI-RS	Cell Specific (2200 4x1) / User Specific
Additional Path Loss (db)	0 / 5 / 10 / 15

AVQC				SBTS24R3_ENB_9999_240318_000007			SBTS24R3_ENB_9999_240318_000007			
				Avg_Dltput	Avg_Dltput	Avg_Dltput	Avg_Dltput	Avg_Dltput	Avg_Dltput	Avg_SRSEpre
Tes	bfSubba	UE	Channel-Model	CB08322 Tput [avg]	CB08247 Tput [avg]	Gain/Loss	CB08322 Tput [avg]	CB08247 Tput [avg]	Gain/Loss	SRSEpre
8	8	3	CDL-C-Umi	1495	1289	15.98	219	221	-0.90	-90.8
9	8	3	CDL-D-Umi	1320	1126	17.23	233	233	0.00	-96.17
10	8	3	CDL-C-Uma	1346	1264	6.49	200	197	1.52	-90.8
11	8	3	CDL-D-Uma	1320	1164	13.40	233	233	0.00	-96
12	8	15	CDL-C-Umi	1240	1208	2.65	215	211	1.90	-94.5
13	8	15	CDL-D-Umi	1051	1067	-1.50	233	233	0.00	-96
14	8	15	CDL-C-Uma	1180	1199	-1.58	197	197	0.00	-89.5
15	8	15	CDL-D-Uma	1064	1139	-6.58	233	233	0.00	-96.19

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